Training 3: Lagrangian duality

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Scheduling of today training lecture
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Together we will set the **sizing problem** of a truss with **five** design variables (cross-sectional areas). Then, using Matlab, **you** will try to solve it using the **Lagrangian duality** approach.

I will stay here to help, if needed!

No reports are required for this lecture.

Class will start at **14:15**.

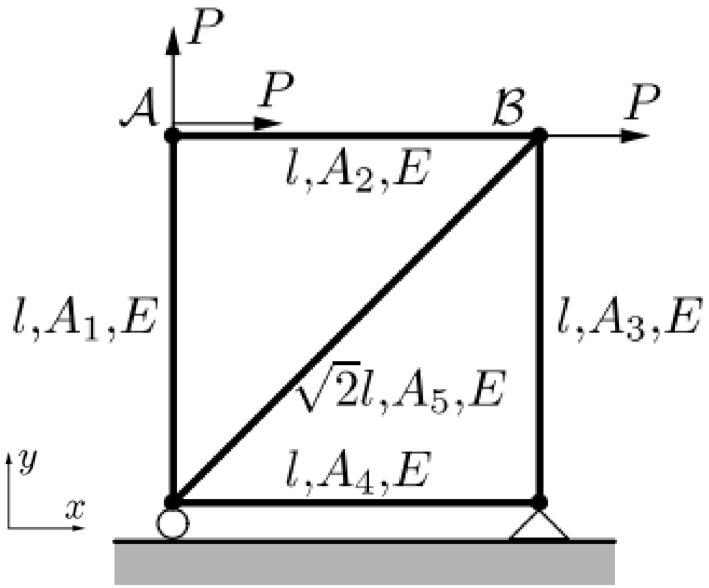
Exercise 1

Exercise 3.8 The volume of the five-bar truss in the Fig. 3.15 should be minimized given that the truss has to be sufficiently stiff. Specifically, the compliance has to be lower than the value c_0. All bars have Young's modulus E. The design variables are the cross-sectional areas of the bars: $A_1, A_2, \ldots, A_5$. The truss is subjected to three forces $P > 0$, so that the compliance may be written as

$$P u_x^{\mathcal{A}} + P u_y^{\mathcal{A}} + P u_x^{\mathcal{B}},$$

where $(u_x^{\mathcal{N}}, u_y^{\mathcal{N}})$ are the displacements of the node $\mathcal{N}$.

a) Formulate the problem as a mathematical programming problem.
b) Solve the optimization problem by using Lagrangian duality.



$$\left\{ \begin{array}{lll} \min\limits_{\boldsymbol{x}} g_0(\boldsymbol{x}) & & \\ \text{s.t.} & g_i(\boldsymbol{x}) \leq 0, & i = 1, \ldots, l \\ & \boldsymbol{x} \in \mathcal{X}, & \end{array} \right.$$

Lagrangian duality

The solution to any constrained problem starts with the definition of the **Lagrangian function**, which is a function that mixes both the objective function and the constraint(s).

$$L = L(\mathbf{x}, \boldsymbol{\lambda}) = f + \sum \lambda_i \, g_i$$

In this problem, the design variables **x** are the five cross-sectional areas, while the constraint is just one (on compliance). Therefore we will look for a Lagrangian of six variables:

$$L = L(A_1, A_2, A_3, A_4, A_5, \lambda)$$

The Lagrangian function is then the tool which allows to solve the constrained problem (minization of the function f). A possible approach is imposing the KKT conditions, but…

Lagrangian duality

…It may be quite tedious to obtain an optimal solution of the problem by solving the nonlinear equations and inequalities constituting the KKT conditions. **There exists another method to obtain an optimal solution that will prove more suitable, especially for large-scale structural optimization problems:**

Dual objective function

$$\varphi(\boldsymbol{\lambda}) = \min_{\boldsymbol{x} \in \mathcal{X}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}).$$

$$\boxed{(\mathbb{D}) \quad \begin{cases} \max\limits_{\boldsymbol{\lambda}} \varphi(\boldsymbol{\lambda}) \\ \text{s.t.} \quad \boldsymbol{\lambda} \geq \mathbf{0}, \end{cases}}$$

It can be demonstrated this reformulation is equivalent to the original problem for **convex problems** in which solutions to the primal and dual problem are the same.

Exercise 1
Returning to the problem

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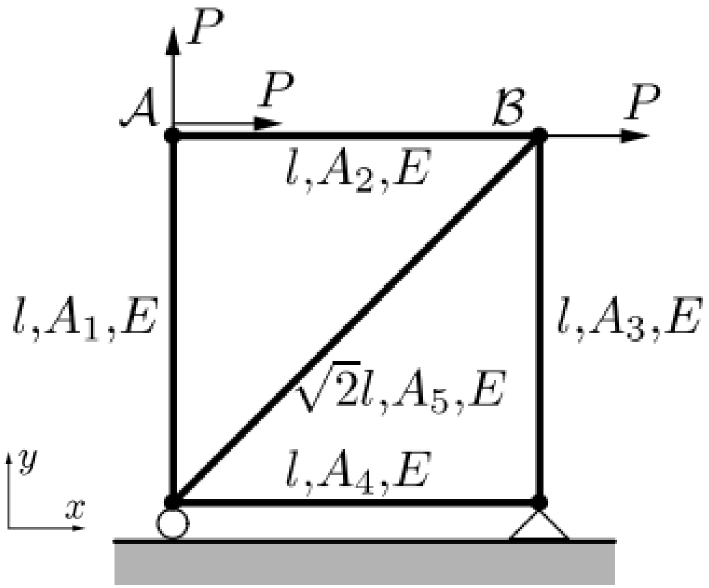
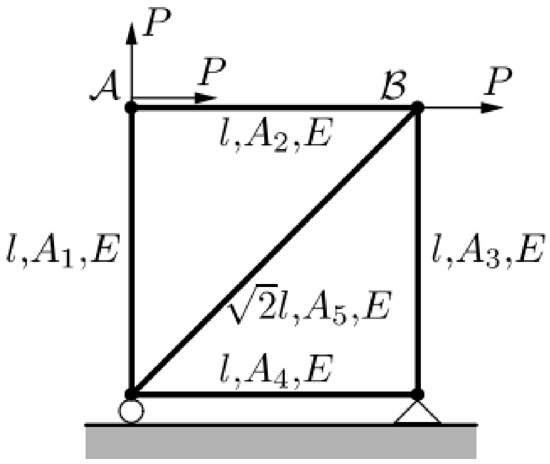


Exercise 1
Hints to solve the problem

The mass of the structure, i.e. the objective function, is:

$$A_1 + A_2 + A_3 + A_4 + \sqrt{2}A_5$$

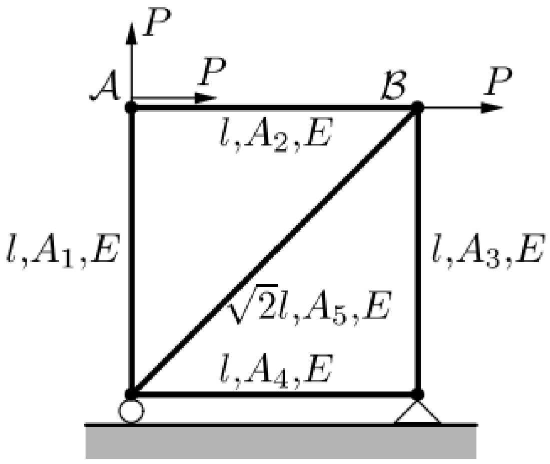
Where parameters ρ and L are neglected or, equivalently, set equal to unity.



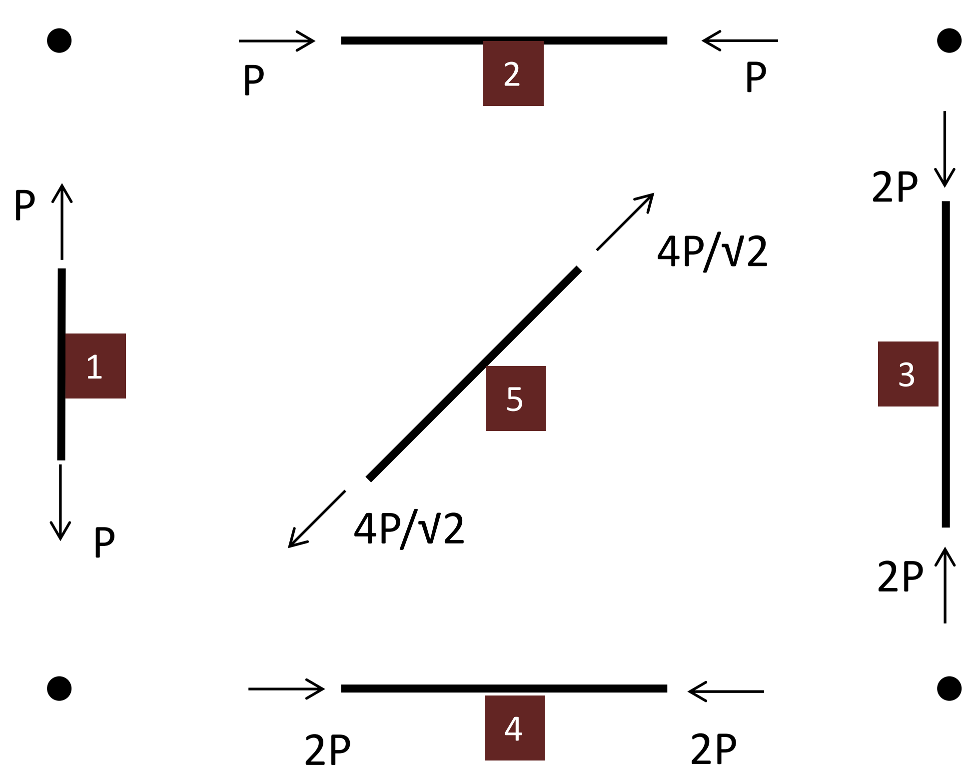
Exercise 1
Hints to solve the problem

To write the constraint on compliance as function of the five design variables (the five cross-sectional areas), the state of stress inside the bars must be found.
The structure is isostatic, therefore there is no need to build the stiffness matrix. The frame can be solved using equilibrium equations only.

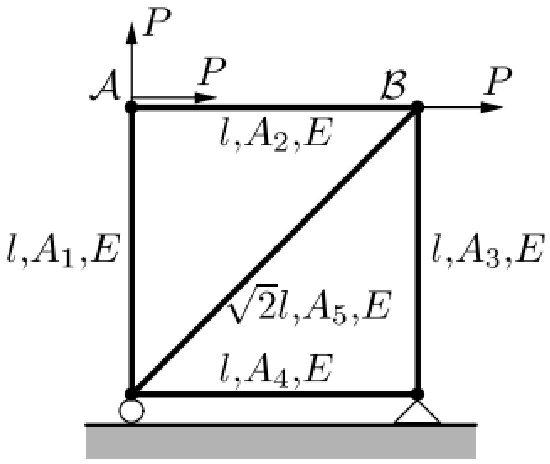
1 – Find the reaction forces of the hinge and roller;
2 – Explode the structure and find internal forces by resolving equilibrium equations.



Exercise 1
Hints to solve the problem



Explode the structure and find internal forces by resolving equilibrium equations…

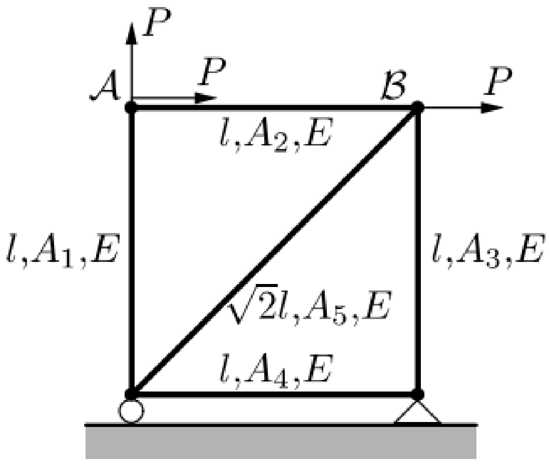


Exercise 1
Hints to solve the problem

Once the force **s** acting inside each bar is known, the stress is easily computed as s/A. Thanks to the way elements are linked one to the other, the state of stress inside each of them is uniaxial. A multiaxial state of stress does not invalidate any of the principles seen with this exercise, but complicates a lot the computations.

For **uniaxial state of stress** (tension or compression) and **linear-elastic material**, the elongation δ of a bar is easily computed as:

$$\delta = \frac{L}{EA} s$$



Exercise 1
Hints to solve the problem

The **compliance** of a structure is its elastic energy of deformation (strain energy).

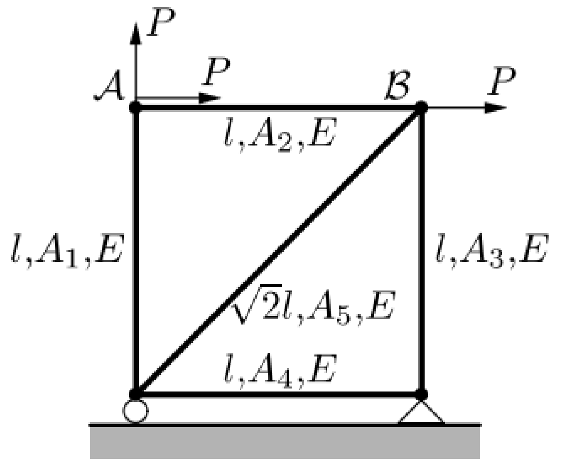
$$\mathrm{C} = \mathbf{F}^{\mathrm{T}}\mathbf{u}$$

$$\boxed{\mathrm{W}_{\mathrm{ext}} = \mathbf{F}^{\mathrm{T}}\mathbf{u} = \mathbf{s}^{\mathrm{T}}\boldsymbol{\delta} = \mathrm{W}_{\mathrm{int}}}$$

Since the nodal displacements **u** are not known yet, it is useful to use the equivalent definition (coming from the conservation of energy):

$$\mathrm{C} = \mathbf{s}^{\mathrm{T}}\boldsymbol{\delta}$$

Which is a straightforward application of the Principle of virtual works.



Exercise 1
Hints to solve the problem

$$\mathbf{s}^{\mathrm{T}} = \left[\mathrm{P}, -\mathrm{P}, -2\mathrm{P}, -2\mathrm{P}, \frac{4\mathrm{P}}{\sqrt{2}}\right]$$

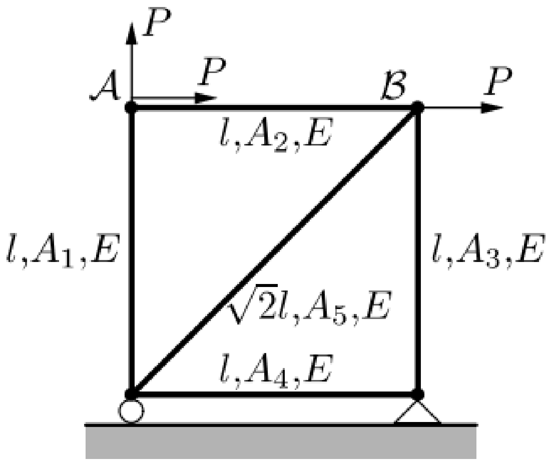
$$\boldsymbol{\delta} = \begin{bmatrix} 1/A_1 \\ -1/A_2 \\ -2/A_3 \\ -2/A_4 \\ 4/A_5 \end{bmatrix} \frac{\mathrm{PL}}{\mathrm{E}}$$

The **compliance** of a structure is then:

$$\mathrm{C} = \mathbf{s}^{\mathrm{T}}\boldsymbol{\delta} = \frac{\mathrm{P}^2\mathrm{L}}{\mathrm{E}}\left(\frac{1}{\mathrm{A}_1} + \frac{1}{\mathrm{A}_2} + \frac{4}{\mathrm{A}_3} + \frac{4}{\mathrm{A}_4} + \frac{16}{\sqrt{2}\mathrm{A}_5}\right)$$

While the constraint is simply:

$$\mathrm{C} \leq \mathrm{C}_0$$

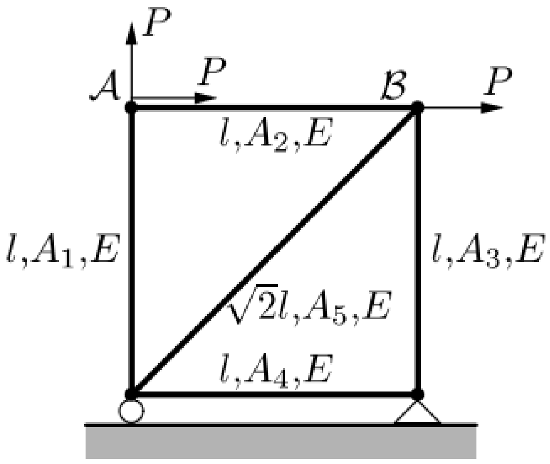


Exercise 1
Hints to solve the problem

The **optimization problem** becomes:

$$
\begin{cases}
\min \quad A_1 + A_2 + A_3 + A_4 + \sqrt{2} A_5 \\
\text{s.t.} \quad \begin{cases} \dfrac{1}{A_1} + \dfrac{1}{A_2} + \dfrac{4}{A_3} + \dfrac{4}{A_4} + \dfrac{8\sqrt{2}}{A_5} - \dfrac{Ec_0}{P^2 l} \leq 0 \\ A_1, \ldots, A_5 \geq 0. \end{cases}
\end{cases}
$$

Note that the problem is **separable**, that is, all the functions defining the problem can be written as a sum of contributions, all containing one D.V. only.



Exercise 1
Solve the problem

Solve the problem stated in the previous slide using the **Lagrangian duality** approach.

1. Write the Lagrangian of the system, $L(A_1, A_2, A_3, A_4, A_5, \lambda)$;
2. Compute the partial derivatives of the Lagrangian w.r.t. the five design variables, $\frac{\partial L}{\partial A_i} = h_i(A_i, \lambda)$;
3. Build the dual function ϕ.

$$\varphi(\boldsymbol{\lambda}) = \min_{\boldsymbol{x} \in \mathcal{X}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}).$$

The dual function represents the locus of the minima of the **Lagrangian** function inside the admissible domain *X*. Since the Lagrangian is composed by a linear combination of convex functions, it is convex too. Therefore, a point with a null gradient is for sure a minimum. To identify the locus of minima, as function of the parameter λ, it is sufficient to evaluate the Lagrangian at the condition grad(L) = 0.

Exercise 1
Solve the problem

3. Build the dual function ϕ.

$$\varphi(\boldsymbol{\lambda}) = \min_{\boldsymbol{x} \in \mathcal{X}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}).$$

The function ϕ is obtained evaluating the Lagrangian at points satisfying:

$$\nabla L = \begin{bmatrix} \frac{\partial L}{\partial A_1} \\ \frac{\partial L}{\partial A_2} \\ \frac{\partial L}{\partial A_3} \\ \frac{\partial L}{\partial A_4} \\ \frac{\partial L}{\partial A_5} \end{bmatrix} = \bar{0}$$

Exercise 1
Solve the problem

4. Solve each $\frac{\partial L}{\partial A_i} = h_i(A_i, \lambda) = 0$ for $A_i = A_i(\lambda)$ and substitute A_i back into L in order to obtain the dual function ϕ;
5. Solve the maximization problem:

$$\boxed{(\mathbb{D}) \qquad \begin{cases} \max\limits_{\boldsymbol{\lambda}} \varphi(\boldsymbol{\lambda}) \\ \text{s.t.} \quad \boldsymbol{\lambda} \geq \mathbf{0}, \end{cases}}$$

Exercise 1
Hints to solve the problem

There is an useful property which links the derivative of the **dual objective function** ϕ to the **active** constraint g_1:

$$\frac{d\varphi}{d\lambda} = g_1[A_1(\lambda), A_2(\lambda), A_3(\lambda), A_4(\lambda), A_5(\lambda)]$$

Where $A_i(\lambda)$ is the *i-th* locus of stationary points of L with respect to A_i. Therefore:

$$\frac{d\varphi(\lambda)}{d\lambda} = \varphi'(\lambda) = 0 \leftrightarrow g_1(\lambda) = 0$$

Exercise 1
Note

In the procedure, some constraints (i.e. **box constraints**) are neglected.

$$A_1, \ldots, A_5 \geq 0.$$

At the end of the solution procedure, one should always check that the solution falls inside the bounds defined for each D.V. In this exercise these box constraints are never active, and are *ipso facto* satisfied: no check is needed.

The general procedure to cope with solutions outside the box is iterative:

I. Set the value of the D.V.s exceeding their limits to the closest ones;
II. Solve again the problem keeping the value of the exceeding D.V.s fixed;
III. Check if the solution found satisfies all the remaining box constraints. If needed, repeat from point I.

Example

$2 \leq x \leq 5$	$x^* = 7$	$\rightarrow$	$x^* = 5$
$0 \leq y \leq 10$	$y^* = 3$		Solve again the problem for the y variable only (x is no more a variable)

Matlab commands

Variable definition

var = sym('var');

Substitution of a value inside a symbolic expression

new_expr = subs(old_expr, old_var, new_var);

Get the list of variables for a symbolic function

var_list = symvar(sym_fun);

Solve a function

x_sol = solve(fun, var, 'returnconditions', true);

Differentiate a function

dfundx = diff(fun, var);

For further help see: [Symbolic Math Toolbox Documentation (mathworks.com)]